A Resolvent-Algebra Version of Algorithm 5.1

Avoiding a full explicit $\mathbf{Q}$-basis for the splitting field

Self-contained notes on the improved cubic-surface algorithm

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Abstract

Algorithm 5.1 of Elsenhans–Jahnel constructs a smooth cubic surface over $\mathbf{Q}$ with prescribed Galois action on its 27 lines. In the direct implementation, one represents the splitting field $L/\mathbf{Q}$ as a $\mathbf{Q}$-vector space, writes the Galois automorphisms as large rational matrices, and solves semilinear descent equations in dimensions 6 and 10. This note describes a more economical version. The ten-dimensional projective descent is unnecessary if one evaluates Clebsch invariants directly on the twisted six-dimensional source. The remaining six-dimensional descent can be modeled using lower-degree resolvent algebras attached to suitable G-sets, especially the 27-line G-set, instead of using the full splitting field. The resulting algorithm takes as arithmetic input a separable polynomial $f(x)$ with labelled Galois group and a homomorphism into $W(E_6)$; it avoids constructing a $\mathbf{Q}$-basis of the full splitting field unless one explicitly chooses the regular G-set.

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1 Purpose and main conclusion

The goal is to construct a smooth cubic surface $X/\mathbf{Q}$ whose induced Galois action on

$$\mathrm{Pic}(X_{\overline{\mathbf{Q}}})$$

— equivalently, on the configuration of the 27 geometric lines — realizes a prescribed representation

$$\rho : G_{\mathbf{Q}} \longrightarrow W(E_6).$$

Computationally, one usually replaces $G_{\mathbf{Q}}$ by a finite quotient. Thus the input is a finite Galois extension $L/\mathbf{Q}$, a finite group $G = \mathrm{Gal}(L/\mathbf{Q})$, and a homomorphism

$$\rho : G \longrightarrow W(E_6).$$

The original descent-heavy implementation represents L explicitly as a $\mathbf{Q}$-vector space. This is often the dominant cost.

The improvement in this note has two parts.

Improvement 1. Do not construct the descended ten-dimensional projective model $\widetilde{M}_\rho \subset \mathbf{P}^9_{\mathbf{Q}}$ unless one specifically wants it. For the purpose of producing a cubic equation, it is enough to evaluate the $W(E_6)$-invariant Clebsch quantities directly on the twisted six-dimensional source.

Improvement 2. Do not construct the full splitting field L as a $\mathbf{Q}$-vector space when building the twisted six-dimensional source. Instead use a lower-degree resolvent algebra associated with a G-set Ω whose permutation module contains the six-dimensional E_6 reflection representation.

The resulting mathematical input is not merely the abstract group G. One needs both:

$$\boxed{\text{a polynomial } f(x) \in \mathbf{Q}[x] \text{ whose splitting field is } L,}$$

and

$$\boxed{\text{a labelled Galois group } G \leq S_{\deg f} \text{ together with } \rho : G \to W(E_6).}$$

The polynomial encodes the arithmetic G-torsor. The labelled permutation group records how this torsor acts on the roots. The homomorphism ρ transfers this arithmetic torsor to the E_6 marking data.

Main conclusion. The full splitting field L is still present mathematically, but it need not be represented as a $\mathbf{Q}$-vector space. It can be replaced by smaller associated étale algebras

$$\mathcal{A}_\Omega = (L \otimes_{\mathbf{Q}} \mathbf{Q}^\Omega)^G$$

attached to carefully chosen G-sets Ω. The universal safe choice is the 27-line G-set obtained by restricting the standard $W(E_6)$-action on the 27 lines along ρ.

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2 The mathematical problem

2.1 Input

The improved algorithm has the following input.

Input 1. A separable polynomial

$$f(x) \in \mathbf{Q}[x]$$

with splitting field L.

Input 2. A labelled Galois group computation: a subgroup

$$G \leq S_d, \qquad d = \deg f,$$

identified with $\mathrm{Gal}(L/\mathbf{Q})$ through its action on an ordered list of the roots

$$\alpha_1, \ldots, \alpha_d.$$

Equivalently, for each $g \in G$, the action is

$$g(\alpha_i) = \alpha_{g(i)}.$$

Input 3. A homomorphism

$$\rho : G \longrightarrow W(E_6).$$

The embedding of the image into $W(E_6)$ is part of the input; the abstract isomorphism type of the image is not enough.

Input 4. Universal precomputed E_6 data:

- the six-dimensional rational representation V_6 of $W(E_6)$ coming from Pinkham's cusp-parameter construction;
- the 40 normalized Coble gamma labels and their convention sign mask;
- the formulas for the $W(E_6)$-invariant Clebsch quantities A, B, C, D, E;
- the 27-line $W(E_6)$-set and the $W(E_6)$-equivariant map from its permutation module onto V_6.

2.2 Output

The desired output is a homogeneous cubic form

$$F(X_0, X_1, X_2, X_3) \in \mathbf{Q}[X_0, X_1, X_2, X_3]$$

such that the surface

$$X = \{F = 0\} \subset \mathbf{P}^3_{\mathbf{Q}}$$

is smooth and the induced Galois representation on the 27 geometric lines of X agrees with ρ, up to the marking convention used to identify the line configuration with the standard E_6 configuration.

2.3 Why an abstract group is insufficient

Two different polynomials can have labelled Galois groups abstractly isomorphic to the same subgroup of $W(E_6)$ but have non-isomorphic splitting fields. They define different classes in

$$H^1(\mathbf{Q}, G),$$

and hence can define different twists of the marked moduli space. The abstract group determines the possible symmetry type; the polynomial determines the actual arithmetic torsor.

Thus the data

$$G \quad \text{and} \quad \rho : G \to W(E_6)$$

are not enough. One also needs the polynomial f, or an equivalent presentation of the same G-torsor over $\mathbf{Q}$.

3 Universal geometry used by the algorithm

3.1 The marked cubic-surface moduli space

Let $\widetilde{\mathcal{M}}$ denote the moduli space of marked smooth cubic surfaces. A marking is an ordered six-tuple of pairwise disjoint lines. After blowing down these six lines, a marked cubic surface over an algebraically closed field is the same as an ordered configuration of six points

$$p_1, \ldots, p_6 \in \mathbf{P}^2$$

in general position, modulo the diagonal action of PGL_3.

The Weyl group $W(E_6)$ acts on the marking. The quotient by this action is the unmarked moduli space. Algorithm 5.1 prescribes a Galois action on the marking, then forgets the marking after computing invariant moduli data.

3.2 The 27 lines and the six-dimensional representation

Use the standard blow-up notation

$$\mathrm{Pic}(X_{\overline{\mathbf{Q}}}) = \mathbf{Z}H \oplus \mathbf{Z}E_1 \oplus \cdots \oplus \mathbf{Z}E_6,$$

with intersection form

$$H^2 = 1, \qquad E_i^2 = -1, \qquad H \cdot E_i = 0, \qquad E_i \cdot E_j = 0 \quad (i \neq j).$$

The canonical class is

$$K = -3H + E_1 + \cdots + E_6.$$

The 27 line classes are

$$E_i, \qquad H - E_i - E_j \quad (1 \leq i < j \leq 6),$$

and

$$2H - \sum_{j \neq i} E_j \quad (1 \leq i \leq 6).$$

Let

$$V_6 = K^\perp \otimes_{\mathbf{Z}} \mathbf{Q}.$$

This is the rational six-dimensional reflection representation of $W(E_6)$.

Let Ω_{27} be the set of the 27 line classes. There is a $W(E_6)$-equivariant surjection

$$q_{27} : \mathbf{Q}^{\Omega_{27}} \longrightarrow V_6 \tag{1}$$

defined on basis vectors by

$$e_\ell \longmapsto \ell + \frac{1}{3}K. \tag{2}$$

Indeed,

$$\left(\ell + \frac{1}{3}K\right) \cdot K = \ell \cdot K + \frac{1}{3}K^2 = -1 + 1 = 0.$$

The images of the 27 line classes span $K^\perp \otimes \mathbf{Q}$.

Implementation point. The map q_{27} is the universal reason that the 27-line resolvent is enough: after restricting the 27-line action along $\rho : G \to W(E_6)$, the associated permutation module still surjects onto the desired six-dimensional source representation.

3.3 Pinkham's cuspidal-cubic source

Let

$$C : yz^2 = x^3 \subset \mathbf{P}^2$$

be the cuspidal cubic. Its smooth locus is parametrized by

$$p(t) = (t : t^3 : 1), \qquad t \in \mathbf{A}^1.$$

For

$$t = (t_1, \ldots, t_6) \in \mathbf{A}^6,$$

put

$$p_i(t) = p(t_i).$$

When these six points are in general position, blowing them up gives a marked smooth cubic surface. Thus one obtains a rational map

$$\mathbf{A}^6 \dashrightarrow \widetilde{\mathcal{M}}.$$

The source $\mathbf{A}^6$ carries Pinkham's six-dimensional $W(E_6)$-action, and the map to $\widetilde{\mathcal{M}}$ is $W(E_6)$-equivariant.

The general-position locus is the complement of the E_6 root hyperplanes. More explicitly, define

$$\boxed{\delta_{E_6}(t) = \left(\prod_{1 \le i < j \le 6} (t_j - t_i)\right) \left(\prod_{1 \le i < j < k \le 6} (t_i + t_j + t_k)\right) (t_1 + \cdots + t_6).} \tag{3}$$

Then the regular locus is

$$V^\circ = \{t \in \mathbf{A}^6 : \delta_{E_6}(t) \neq 0\}.$$

The factors have the following meanings:

- $t_i = t_j$: two of the six points coincide;
- $t_i + t_j + t_k = 0$: three of the six points are collinear;
- $t_1 + \cdots + t_6 = 0$: all six points lie on a conic.

Important search-domain rule. One should not exclude $t_i = 0$. The point $p(0) = (0 : 0 : 1)$ is smooth on the cuspidal cubic. The only geometric exclusions in the source are the factors of δ_{E_6} in (3), plus later Clebsch, pentahedral, and smoothness failure loci.

For invariant boundary testing it is often preferable to use

$$\Delta_{E_6}(t) = \delta_{E_6}(t)^2. \tag{4}$$

The polynomial Δ_{E_6} is $W(E_6)$-invariant, whereas δ_{E_6} is a relative invariant.

4 Coble coordinates and Clebsch invariants

4.1 Normalized Coble coordinates on the cusp

For $I \subseteq \{1, \ldots, 6\}$, write

$$s_I(t) = \sum_{i \in I} t_i, \qquad s_{[6]}(t) = t_1 + \cdots + t_6,$$

and

$$\Delta_I(t) = \prod_{\substack{i<j \\ i,j \in I}} (t_j - t_i), \qquad \Delta_{\mathrm{VdM}}(t) = \Delta_{\{1,\ldots,6\}}(t).$$

On the cuspidal family, every raw Coble gamma coordinate has a common Vandermonde factor Δ_{VdM}. Dividing by this factor gives normalized degree-nine coordinates.

For a decomposition

$$I \sqcup I^c = \{1, \ldots, 6\}, \qquad |I| = |I^c| = 3,$$

the type-I normalized coordinate is

$$\boxed{\widehat{\gamma}_{I|I^c}(t) = \Delta_I(t)\Delta_{I^c}(t)s_I(t)s_{I^c}(t)s_{[6]}(t).} \tag{5}$$

There are 10 such coordinates.

For a partition of $\{1, \ldots, 6\}$ into three unordered pairs A, B, C, and a cyclic ordering (A, B, C) of the three pairs, define

$$d_A = t_{a_2} - t_{a_1} \quad \text{for } A = \{a_1 < a_2\},$$

and similarly for B and C. The type-II normalized coordinate is

$$\boxed{\widehat{\gamma}_{A \to B \to C}(t) = d_A d_B d_C \prod_{a \in A} s_{\{a\} \cup B} \prod_{b \in B} s_{\{b\} \cup C} \prod_{c \in C} s_{\{c\} \cup A}.} \tag{6}$$

There are 30 such coordinates, because reversing the cyclic orientation gives the other coordinate associated with the same three-pair partition.

Choose once and for all an ordering of the 40 labels and a sign mask

$$\eta = (\eta_1, \dots, \eta_{40}) \in \{\pm 1\}^{40}$$

matching the implementation convention for Coble's determinants. The stored normalized gamma vector is

$$\widehat{\Gamma}(t) = \left(\eta_1 \widehat{\gamma}_1(t), \dots, \eta_{40} \widehat{\gamma}_{40}(t)\right)^t.$$

With the correct sign mask, this vector is exactly equivariant for a signed permutation representation of $W(E_6)$:

$$\widehat{\Gamma}(M_w t) = \widehat{P}_w \widehat{\Gamma}(t), \qquad w \in W(E_6), \tag{7}$$

where M_w denotes Pinkham's six-dimensional matrix and $\widehat{P}_w$ is a signed permutation matrix on the 40 normalized gamma labels.

Implementation point. The sign mask is not cosmetic. A mismatch here usually appears later as a failure of rationality for supposedly descended quantities. The even power sums below are independent of changing the signs of individual gamma coordinates, but equivariance checks and comparisons with stored Coble data are not.

4.2 Clebsch invariants

Let

$$P_k(t) = \sum_{a=1}^{40} \widehat{\gamma}_a(t)^k.$$

The sum is over one representative from each signed pair of Coble coordinates. In the formulas below only even powers appear, so the chosen signs of the forty coordinates do not affect the value. One should not sum over all eighty signed coordinates, since that introduces an unwanted factor of two.

The Clebsch quantities are

$$\begin{aligned}
A &= -6P_2, \\
B &= -24P_4 + \frac{41}{16}P_2^2, \\
C &= \frac{576}{13}P_6 - \frac{396}{13}P_4P_2 + \frac{29}{13}P_2^3, \\
D &= -\frac{62208}{1171}P_8 + \frac{54864}{1171}P_6P_2 + \frac{203616}{1171}P_4^2 - \frac{61287}{1171}P_4P_2^2 + \frac{13393}{4684}P_2^4, \\
E &= \frac{41472}{155}P_{10} - \frac{4605984}{36301}P_8P_2 - \frac{106272}{403}P_6P_4 + \frac{19990440}{471913}P_6P_2^2 \\
&\quad + \frac{47719206}{471913}P_4^2P_2 - \frac{7468023}{471913}P_4P_2^3 + \frac{10108327}{18876520}P_2^5.
\end{aligned} \tag{8}$$

They are $W(E_6)$-invariant polynomial functions on the six-dimensional source. Consequently they descend to rational functions, indeed polynomial functions, on any twist of that source.

If the gamma vector is multiplied by a scalar λ, then

$$(A, B, C, D, E) \mapsto (\lambda^2 A, \lambda^4 B, \lambda^6 C, \lambda^8 D, \lambda^{10} E).$$

Thus the moduli point lies in the weighted projective space

$$\mathbf{P}(1, 2, 3, 4, 5).$$

4.3 From Clebsch invariants to a cubic equation

A general smooth cubic surface has a Sylvester pentahedral form

$$\begin{cases} a_0W_0^3 + a_1W_1^3 + a_2W_2^3 + a_3W_3^3 + a_4W_4^3 = 0, \\ W_0 + W_1 + W_2 + W_3 + W_4 = 0. \end{cases} \tag{9}$$

Let

$$\sigma_i = e_i(a_0, \dots, a_4), \qquad 1 \le i \le 5.$$

Given a Clebsch vector $[A : B : C : D : E]$ with $E \neq 0$, use

$$\boxed{[\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5] = \left[B,\ D,\ \frac{C^2 - AE}{4},\ CE,\ E^2\right].} \tag{10}$$

Then form the monic quintic

$$g(T) = T^5 - \sigma_1 T^4 + \sigma_2 T^3 - \sigma_3 T^2 + \sigma_4 T - \sigma_5. \tag{11}$$

Let

$$\mathcal{A}_5 = \mathbf{Q}[T]/(g), \qquad \theta = T \bmod g.$$

Assume g is separable. Let

$$N = \ker\bigl(\mathrm{Tr}_{\mathcal{A}_5/\mathbf{Q}} : \mathcal{A}_5 \to \mathbf{Q}\bigr).$$

Choose a $\mathbf{Q}$-basis

$$c_0, c_1, c_2, c_3$$

of N and set

$$\ell(X) = c_0X_0 + c_1X_1 + c_2X_2 + c_3X_3 \in \mathcal{A}_5[X_0, X_1, X_2, X_3].$$

The output cubic is

$$\boxed{F(X_0, X_1, X_2, X_3) = \mathrm{Tr}_{\mathcal{A}_5/\mathbf{Q}}\bigl(\theta\, \ell(X)^3\bigr).} \tag{12}$$

This is the trace-algebra form of Algorithm A.10. It avoids computing the splitting field of the pentahedral quintic.

5 What the original descent computes

Let $L/\mathbf{Q}$ be the splitting field of f and let $G = \mathrm{Gal}(L/\mathbf{Q})$. The six-dimensional twisted source is

$$V_\rho = (L \otimes_{\mathbf{Q}} V_6)^G, \tag{13}$$

where G acts semilinearly by

$$g(a \otimes v) = g(a) \otimes \rho(g)v.$$

In coordinates, a point of V_ρ is a vector $t \in L^6$ satisfying equations of the form

$$M_{\rho(g)}\, g(t) = t, \qquad g \in G, \tag{14}$$

up to the chosen row/column convention. The original implementation constructs a basis matrix

$$D_\rho \in \mathrm{GL}_6(L)$$

whose columns form a $\mathbf{Q}$-basis of this fixed space.

Similarly, the ten-dimensional Coble span has a G-twist. The original implementation constructs a basis matrix

$$B_\rho \in \mathrm{GL}_{10}(L)$$

for the ten-dimensional semilinear fixed space, descends the thirty Coble cubics to $\mathbf{Q}$, searches for a rational point in the descended projective variety, and then maps the point back to Coble coordinates.

To solve (14) and its ten-dimensional analogue by plain rational linear algebra, one often chooses a $\mathbf{Q}$-basis

$$\omega_1, \dots, \omega_n$$

of L and writes every field automorphism $g \in G$ as a matrix in $\mathrm{GL}_n(\mathbf{Q})$. This is the step we want to avoid. When $|G| = [L : \mathbf{Q}]$ is large, the matrices are large, the field arithmetic is expensive, and the descent becomes the bottleneck.

The full splitting field is one model of the torsor, not the only model. The algorithm needs the twist V_ρ, not a particular $\mathbf{Q}$-basis of L. Resolvent algebras provide smaller models of the same torsor whenever the desired representation is visible inside a low-degree permutation representation.

6 First simplification: eliminate the ten-dimensional descent

The ten-dimensional matrix B_ρ is useful if one wants explicit descended coordinates on the projective Coble model

$$\widetilde{M}_\rho \subset \mathbf{P}^9_{\mathbf{Q}}.$$

It is not needed to produce a cubic equation.

Indeed, the trace cubic (12) depends only on the unmarked Clebsch vector

$$[A : B : C : D : E] \in \mathbf{P}(1, 2, 3, 4, 5)(\mathbf{Q}).$$

The functions A, B, C, D, E in (8) are $W(E_6)$-invariant functions on the Pinkham source V_6. Therefore they descend directly to functions

$$A_\rho, B_\rho, C_\rho, D_\rho, E_\rho$$

on the twisted source V_ρ.

Thus the production algorithm can follow the shorter route

$$u \in V_\rho(\mathbf{Q}) \longmapsto \big(A_\rho(u), B_\rho(u), C_\rho(u), D_\rho(u), E_\rho(u)\big) \longmapsto \text{trace cubic over } \mathbf{Q}.$$

No rational point search in $\mathbf{P}^9$, no descended Coble cubics, and no B_ρ are required.

Proposition 6.1 (Ten-dimensional descent is unnecessary for equation production)**.** *Assume one can enumerate rational points of the twisted source V_ρ and evaluate the descended Clebsch functions $A_\rho, B_\rho, C_\rho, D_\rho, E_\rho$ on those points. Then one can run the cubic-surface production part of Algorithm 5.1 without constructing the ten-dimensional matrix B_ρ or the descended projective model $\widetilde{M}_\rho \subset \mathbf{P}^9_{\mathbf{Q}}$.*

Proof. Over L, a point of V_ρ becomes an ordinary cusp-parameter vector $t \in V_6$. The normalized Coble map gives a marked cubic surface, and the power-sum formulas (8) give its unmarked Clebsch invariants. Since these formulas are $W(E_6)$-invariant, the resulting weighted Clebsch point is fixed by Galois and is defined over $\mathbf{Q}$. Formula (10) and the trace construction (12) then produce a cubic equation over $\mathbf{Q}$. The intermediate descended Coble coordinates are only one possible way of obtaining the same Clebsch point. □

7 Resolvent algebras attached to G-sets

7.1 The associated étale algebra

Let Ω be a finite left G-set. Write $\mathbf{Q}^\Omega$ for the permutation module with basis $\{e_\omega : \omega \in \Omega\}$. Define

$$\mathcal{A}_\Omega = (L \otimes_{\mathbf{Q}} \mathbf{Q}^\Omega)^G, \tag{15}$$

where G acts semilinearly on L and by its given permutation action on $\mathbf{Q}^\Omega$.

Then $\mathcal{A}_\Omega$ is an étale $\mathbf{Q}$-algebra of degree $|\Omega|$. After base change to L, it splits canonically as

$$L \otimes_{\mathbf{Q}} \mathcal{A}_\Omega \cong L^\Omega. \tag{16}$$

If Ω is transitive and $\Omega \cong G/H$, then

$$\mathcal{A}_\Omega \cong L^H. \tag{17}$$

For a general G-set decomposed into orbits

$$\Omega = \Omega_1 \sqcup \cdots \sqcup \Omega_r, \qquad \Omega_j \cong G/H_j,$$

one has

$$\mathcal{A}_\Omega \cong \prod_{j=1}^{r} L^{H_j}. \tag{18}$$

7.2 Constructing $\mathcal{A}_\Omega$ from f

Suppose first that $\Omega \cong G/H$ is transitive. Choose an expression

$$\theta_H = \Theta(\alpha_1, \dots, \alpha_d) \in L$$

which is fixed by H and whose stabilizer in G is exactly H. Then the resolvent polynomial

$$h_H(T) = \prod_{gH \in G/H} (T - g(\theta_H)) \tag{19}$$

has coefficients in $\mathbf{Q}$ and degree $[G : H] = |\Omega|$. For a generic enough choice of Θ, it is separable and

$$\mathbf{Q}[T]/(h_H) \cong L^H \cong \mathcal{A}_\Omega.$$

For a nontransitive G-set, construct one such resolvent for each orbit and take the product algebra. In practice, one chooses Θ as a suitably generic linear or low-degree expression in the roots of f and verifies that the computed resolvent has the expected degree and discriminant nonzero.

Implementation point. A list of separate orbit resolvent polynomials gives the underlying product algebra $\prod_j L^{H_j}$, but some algorithms also need the descended maps induced by G-equivariant linear maps between permutation modules. Those maps encode the relative labelling of the different orbits. Thus the true computational object is not only the product algebra $\mathcal{A}_\Omega$, but $\mathcal{A}_\Omega$ together with the descended endomorphisms and polynomial functions required below.

7.3 Why this avoids the full splitting field

The full splitting field corresponds to the regular G-set G itself:

$$(L \otimes_{\mathbf{Q}} \mathbf{Q}^G)^G \cong L.$$

Using the regular G-set therefore gives degree $|G|$ arithmetic, which is exactly what we are trying to avoid.

A smaller G-set Ω gives degree $|\Omega|$ arithmetic. The 27-line G-set always has total degree 27, often split into smaller orbit factors. If $|G|$ is large, this can be dramatically cheaper than computing in L.

8 Constructing the twisted six-dimensional source from a resolvent

8.1 General representation-theoretic construction

Let V be the six-dimensional representation V_6 of G obtained by restricting Pinkham's $W(E_6)$ representation along

$$\rho : G \to W(E_6).$$

Choose a finite G-set Ω and a G-equivariant surjection

$$q : \mathbf{Q}^\Omega \twoheadrightarrow V. \tag{20}$$

Because $\mathbf{Q}[G]$ is semisimple, q admits a G-equivariant section

$$s : V \hookrightarrow \mathbf{Q}^\Omega. \tag{21}$$

Set

$$e = sq \in \mathrm{End}_G(\mathbf{Q}^\Omega). \tag{22}$$

Then e is a G-equivariant idempotent and

$$\mathrm{im}(e) \cong V.$$

Twisting by the G-torsor defined by f gives a $\mathbf{Q}$-linear idempotent

$$e_\rho : \mathcal{A}_\Omega \longrightarrow \mathcal{A}_\Omega. \tag{23}$$

Its image is the desired six-dimensional twisted source.

Proposition 8.1 (Resolvent model of the twisted source)**.** *With notation as above, there is a canonical isomorphism*

$$\mathrm{im}(e_\rho) \cong (L \otimes_{\mathbf{Q}} V)^G = V_\rho. \tag{24}$$

In particular, $\mathrm{im}(e_\rho)$ *is a six-dimensional* $\mathbf{Q}$*-vector space.*

Proof. The exact sequence

$$0 \longrightarrow \ker(q) \longrightarrow \mathbf{Q}^{\Omega} \xrightarrow{q} V \longrightarrow 0$$

splits G-equivariantly by s. Tensoring with L and taking semilinear G-invariants gives

$$(L \otimes \mathbf{Q}^{\Omega})^G \cong (L \otimes \ker(q))^G \oplus (L \otimes V)^G.$$

Under the identification

$$(L \otimes \mathbf{Q}^{\Omega})^G = \mathcal{A}_{\Omega},$$

the projection onto the second summand is precisely the descended idempotent e_ρ. Therefore its image is V_ρ. □

8.2 How to compute the idempotent

Suppose the matrices for the action of G on $\mathbf{Q}^{\Omega}$ are P_g, and the matrices for the action on V are M_g. With column vectors, equivariance of q means

$$M_g q = q P_g. \tag{25}$$

Choose any rational section s_0 of q, so that $qs_0 = I_V$. Then

$$s = \frac{1}{|G|} \sum_{g \in G} P_g^{-1} s_0 M_g \tag{26}$$

is a G-equivariant section. Indeed,

$$qs = I_V, \qquad P_h s = s M_h \quad (h \in G).$$

Then

$$e = sq$$

is the desired rational idempotent on $\mathbf{Q}^{\Omega}$.

The descended map e_ρ on $\mathcal{A}_{\Omega}$ is the unique $\mathbf{Q}$-linear map whose base change to L is the coordinate-wise map induced by e on L^{Ω}. Implementation methods include:

1. constructing the corresponding resolvent correspondences between the orbit factors L^{H_i} and L^{H_j};
2. computing the map modulo enough good split primes, where $\mathcal{A}_{\Omega} \otimes \mathbf{F}_p \cong \mathbf{F}_p^{\Omega}$, and reconstructing the rational matrix;
3. in a hybrid test implementation, using the full splitting field only to calibrate e_ρ once, then replacing it by the lower-degree resolvent model.

The first two methods avoid constructing a $\mathbf{Q}$-basis of L.

9 Choosing the G-set Ω

9.1 The root set of f

The cheapest possible choice is the root set of f:

$$\Omega_f = \{1, \dots, d\},$$

with its labelled G-action. Then

$$\mathcal{A}_{\Omega_f} = \mathbf{Q}[x]/(f)$$

if f is irreducible; more generally it is the product of the root-algebra factors corresponding to the G-orbits on the roots.

However, this works only if the permutation module $\mathbf{Q}^{\Omega_f}$ contains the six-dimensional source representation as a quotient. A preliminary character test is

$$m = \left\langle \chi_V, \chi_{\Omega_f} \right\rangle_G = \frac{1}{|G|} \sum_{g \in G} \chi_V(g) \cdot \#\operatorname{Fix}_{\Omega_f}(g). \tag{27}$$

If $m = 0$, then $\mathbf{Q}^{\Omega_f}$ cannot contain V.

If $m > 0$, this is still only a preliminary test, because V may be reducible after restriction to G. The definitive test is to solve the linear equations

$$M_g C = C P_g, \qquad g \in G, \tag{28}$$

for matrices

$$C \in \operatorname{Mat}_{6 \times d}(\mathbf{Q}),$$

and check whether some solution has rank 6. Such a solution is precisely a G-equivariant surjection

$$\mathbf{Q}^{\Omega_f} \twoheadrightarrow V.$$

9.2 The 27-line set

The universal safe choice is

$$\Omega = \Omega_{27},$$

the 27-line set, with G-action through ρ. The surjection is the map q_{27} in (1). Therefore the rank test is automatic.

The associated algebra is

$$\mathcal{A}_{27} = (L \otimes \mathbf{Q}^{\Omega_{27}})^G.$$

It has total degree 27. If the image of G has orbits

$$\Omega_{27} = \Omega_1 \sqcup \cdots \sqcup \Omega_r,$$

then

$$\mathcal{A}_{27} \cong \prod_{j=1}^{r} L^{H_j}, \qquad H_j = \operatorname{Stab}_G(\omega_j).$$

Even when the product has several factors, the descended idempotent e_ρ is still defined on the product algebra and has six-dimensional image.

Recommended choice. Try the root algebra $\mathbf{Q}[x]/(f)$ first. If it contains the six-dimensional source representation, it is likely the fastest model. If it does not, use the 27-line resolvent algebra. This gives a uniform degree-27 substitute for the full splitting field.

9.3 A warning about multiple orbits

When Ω has several G-orbits, the underlying algebra

$$\prod_j L^{H_j}$$

by itself is not always enough for implementation. A G-equivariant map

$$\mathbf{Q}^{\Omega} \to V$$

may mix information from different orbits. To descend such a map, one needs the corresponding rational linear maps between the orbit resolvent factors, or an equivalent method that preserves their common origin from the same G-torsor.

Using a single transitive G-set whose permutation representation surjects onto V is technically simpler. But the 27-line set may decompose after restriction to a subgroup $G \subset W(E_6)$, and the product-plus-correspondences model is then the correct object.

10 Evaluating invariant functions without L

Let

$$F \in \mathbf{Q}[V]^G$$

be a G-invariant polynomial function on the six-dimensional representation. In applications, F will be one of

$$\Delta_{E_6}, \qquad A, B, C, D, E.$$

Choose a G-equivariant surjection

$$q : \mathbf{Q}^{\Omega} \twoheadrightarrow V.$$

Then

$$F \circ q \in \mathbf{Q}[\mathbf{Q}^{\Omega}]^G.$$

Twisting by the torsor defined by f gives an ordinary polynomial function

$$F_{\rho,\Omega} : \mathcal{A}_{\Omega} \longrightarrow \mathbf{Q}. \tag{29}$$

Restricting this function to

$$\mathrm{im}(e_\rho) \cong V_\rho$$

gives the descended function on the twisted source.

Equivalently, if $a \in \mathrm{im}(e_\rho)$ and one base changes to L, then a becomes a vector in L^{Ω}. Applying q gives an element of $L \otimes V$, and evaluating F gives an element of L. Because F is G-invariant and a is descended, that element is fixed by G, hence lies in $\mathbf{Q}$. This rational number is $F_{\rho,\Omega}(a)$.

Proposition 10.1 (Invariant evaluation)**.** *The values*

$$\Delta_{E_6,\rho}(a), \quad A_\rho(a), \quad B_\rho(a), \quad C_\rho(a), \quad D_\rho(a), \quad E_\rho(a)$$

for $a \in V_\rho(\mathbf{Q})$ can be computed from the resolvent model $\mathcal{A}_\Omega$ and the descended idempotent e_ρ, without computing a $\mathbf{Q}$-basis of the full splitting field L.

Proof. The construction above defines each value as the evaluation of a descended polynomial function on a finite-dimensional $\mathbf{Q}$-vector space. The data needed to write this polynomial are the resolvent algebra $\mathcal{A}_\Omega$ and the descended linear maps attached to q and e. The full splitting field appears only after base change, where it is used to prove that the construction agrees with the ordinary untwisted invariant F. □

10.1 Practical exact evaluation methods

There are several implementation strategies.

Method 1. **Symbolic descent.** Fix a $\mathbf{Q}$-basis of $\mathcal{A}_\Omega$, write a general element of $\operatorname{im}(e_\rho)$ in six coordinates, and descend the polynomial $F \circ q$ to a polynomial in those six coordinates. This is a one-time instance-specific computation.

Method 2. **Modular split-prime evaluation.** For good primes p at which the relevant resolvent algebras split and all denominators are invertible, identify

$$\mathcal{A}_\Omega \otimes \mathbf{F}_p \cong \mathbf{F}_p^\Omega.$$

Evaluate e, q, and the invariant F over $\mathbf{F}_p$. Because the result is invariant under the residual labelling ambiguity, it is a well-defined value modulo p. Repeat for enough primes and reconstruct the rational value by CRT and rational reconstruction. If Ω has several orbits, the labellings of the factors must be correlated through the same torsor, or the descended correspondences must be used.

Method 3. **Hybrid calibration.** During development, compute the full splitting field for small examples, verify the resolvent evaluation against the direct L-evaluation, and then remove the full splitting-field step in production.

The crucial point is that the individual gamma coordinates are not the intended output of the no-L implementation. The intended output of the source evaluation is the invariant vector

$$[A : B : C : D : E] \in \mathbf{P}(1, 2, 3, 4, 5)(\mathbf{Q}).$$

11 The improved algorithm

11.1 Universal precomputation

Compute once and store the following data.

U1. A concrete rational model of the six-dimensional representation V_6 of $W(E_6)$, with matrices M_w or a word-evaluation routine for the chosen generators of $W(E_6)$.

U2. The 27-line $W(E_6)$-set and the surjection

$$q_{27} : \mathbf{Q}^{\Omega_{27}} \twoheadrightarrow V_6, \qquad e_\ell \mapsto \ell + K/3.$$

U3. The normalized gamma formulas (5) and (6), the ordering of the forty labels, and the convention sign mask.

U4. The invariant formulas

$$\Delta_{E_6} = \delta_{E_6}^2, \qquad A, B, C, D, E.$$

In a direct invariant implementation, one may store these as polynomial functions on V_6 rather than as gamma-coordinate routines.

U5. The pentahedral reconstruction formulas (10), (11), and (12).

11.2 Instance-specific computation

Given $f(x)$, a labelled Galois group $G \leq S_d$, and $\rho : G \to W(E_6)$, proceed as follows.

Step 1. **Restrict the source representation.** Evaluate $\rho(g)$ in the six-dimensional representation V_6 for generators of G. This gives matrices M_g for the G-action on V.

Step 2. **Choose a low-degree permutation model.** First test the root set Ω_f using the intertwiner equations (28). If there is a rank-six equivariant map

$$\mathbf{Q}^{\Omega_f} \twoheadrightarrow V,$$

use $\Omega = \Omega_f$. Otherwise use $\Omega = \Omega_{27}$ with action through ρ and $q = q_{27}$.

Step 3. **Construct the resolvent algebra.** Construct

$$\mathcal{A}_\Omega = (L \otimes \mathbf{Q}^\Omega)^G$$

from f using resolvent polynomials for the orbits of Ω. Do not construct a $\mathbf{Q}$-basis of L.

Step 4. **Construct the descended source.** Choose a G-equivariant surjection

$$q : \mathbf{Q}^\Omega \twoheadrightarrow V$$

and a G-equivariant section s. Form the idempotent

$$e = sq.$$

Descend it to

$$e_\rho : \mathcal{A}_\Omega \to \mathcal{A}_\Omega.$$

Compute a $\mathbf{Q}$-basis

$$b_1, \ldots, b_6$$

for

$$\mathrm{im}(e_\rho) \cong V_\rho.$$

Optionally replace this basis by an LLL-reduced basis for a chosen integral lattice in $\mathrm{im}(e_\rho)$.

Step 5. Enumerate source points. Enumerate primitive vectors

$$u = (u_1, \dots, u_6) \in \mathbf{Z}^6$$

by increasing height and set

$$a(u) = u_1 b_1 + \cdots + u_6 b_6 \in \operatorname{im}(e_\rho).$$

Primitive enumeration is natural because the Coble and Clebsch constructions are homogeneous, so scalar multiples usually give the same moduli point.

Step 6. Boundary test. Evaluate

$$\Delta_{E_6,\rho}(a(u)).$$

If this is zero, reject u and continue. This is the invariant version of the condition that no factor in (3) vanish after base change to L.

Step 7. Evaluate Clebsch invariants. Evaluate

$$A = A_\rho(a(u)), \quad B = B_\rho(a(u)), \quad C = C_\rho(a(u)), \quad D = D_\rho(a(u)), \quad E = E_\rho(a(u)).$$

If $[A : B : C : D : E]$ is undefined or $E = 0$, reject u and continue.

Step 8. Construct the pentahedral quintic. Use (10) to form

$$\sigma_1, \dots, \sigma_5$$

and then form the quintic $g(T)$ in (11). If g is not separable, reject u and continue.

Step 9. Run the trace construction. Construct

$$\mathcal{A}_5 = \mathbf{Q}[T]/(g),$$

choose a basis of the trace-zero subspace, and output the cubic form

$$F(X) = \operatorname{Tr}_{\mathcal{A}_5/\mathbf{Q}}(\theta \ell(X)^3).$$

Step 10. Certify smoothness. Check that the resulting cubic surface is smooth. If it is singular, reject u and continue. Otherwise return F.

11.3 Language-neutral pseudocode

```
function CubicSurfaceFromResolvent(f, labelled_G, rho, universal_data):
    # Input:
    # f: separable polynomial over Q
    # labelled_G <= S_deg(f), identified with Gal(f)
    # rho: labelled_G -> W(E6)
    # Output:
    # smooth cubic surface over Q realizing rho, if a candidate is found

    V_action = restrict_Pinkham_representation(rho)

    if root_permutation_module_has_rank_6_quotient(f, labelled_G, V_action):
```

```
        Omega = root_set_of_f
        q = choose_rank_6_intertwiner(Q^Omega -> V)
    else:
        Omega = line_27_G_set_via_rho
        q = q_27

    A_Omega = construct_resolvent_algebra(f, labelled_G, Omega)

    s = equivariant_section(q)
    e = s * q
    e_rho = descend_idempotent_to_resolvent_algebra(e, A_Omega)

    basis = basis_of_image(e_rho) # six vectors b_1,...,b_6
    basis = optional_LLL_reduce(basis)

    for u in primitive_integer_vectors_by_height(6):
        a = sum(u[i] * basis[i] for i in 1..6)

        if descended_Delta_E(a) == 0:
            continue

        A, B, C, D, E = descended_Clebsch_values(a)
        if weighted_point_is_invalid(A, B, C, D, E):
            continue
        if E == 0:
            continue

        sigmas = [B, D, (C^2 - A*E)/4, C*E, E^2]
        g = T^5 - sigmas[1]*T^4 + sigmas[2]*T^3 \
              - sigmas[3]*T^2 + sigmas[4]*T - sigmas[5]

        if not is_separable(g):
            continue

        F = trace_cubic_from_quintic(g)

        if is_smooth_cubic_surface(F):
            return F

    error("No candidate found in the searched height range")
```

12 Correctness

Theorem 12.1 (Correctness of the improved algorithm)**.** *Assume the universal data use compatible conventions for the $W(E_6)$ action on V_6, the 27 lines, and the normalized Coble gamma coordinates. Assume also that the returned candidate lies in the standard open locus where the Clebsch–pentahedral reconstruction used in Algorithm A.10 recovers the unmarked cubic surface represented by the Clebsch point. If the algorithm in section 11 returns a cubic form F, then the surface $X = \{F = 0\} \subset \mathbf{P}^3_{\mathbf{Q}}$ is a smooth cubic surface whose Galois action on the 27 geometric lines is the prescribed representation ρ, up to the initial marking convention. The optional final computation of the 27 lines certifies this* *conclusion a posteriori.*

Proof. The source space used by the algorithm is

$$\mathrm{im}(e_\rho) \cong V_\rho = (L \otimes V)^G$$

by equation (24). After base change to L, a rational source point becomes an ordinary vector $t \in V$ satisfying the twisted compatibility condition with respect to ρ. The nonvanishing of $\Delta_{E_6,\rho}$ means that t avoids the E_6 root hyperplanes, so the six cuspidal points are in general position and determine a marked smooth cubic surface over L.

The descent condition on t says precisely that applying a Galois element $g \in G$ to the marked surface is the same as changing the marking by $\rho(g) \in W(E_6)$. Hence the associated Galois action on the marked line configuration is the one prescribed by ρ.

The Clebsch quantities A, B, C, D, E are $W(E_6)$-invariant. Therefore their values on the twisted source point are fixed by G and lie in $\mathbf{Q}$. These values are the unmarked moduli invariants of the cubic surface obtained over L.

The pentahedral and trace construction from (10)–(12) constructs a cubic surface over $\mathbf{Q}$ with the same Clebsch invariants. On the open locus where the pentahedral construction is valid and the final smoothness check passes, this surface is the desired unmarked cubic surface. Since the marked object over L descends with cocycle ρ, its line action is the prescribed one, up to the fixed convention identifying markings with the standard E_6 configuration. □

Proposition 12.2 (Termination of exhaustive search)**.** *Assume the enumeration of primitive integer vectors in the chosen source lattice is exhaustive by height. Then, outside the proper failure loci of the Coble/Clebsch and pentahedral constructions, the search eventually finds a valid point.*

Proof. Over L, the source map

$$\mathbf{A}^6 \dashrightarrow \widetilde{\mathcal{M}}$$

coming from six points on the cuspidal cubic is dominant. Twisting by the cocycle ρ gives a dominant rational map

$$V_\rho \dashrightarrow \widetilde{\mathcal{M}}_\rho.$$

The source V_ρ is a six-dimensional vector space over $\mathbf{Q}$, so its rational points are Zariski dense. The boundary $\Delta_{E_6} = 0$, the locus $E = 0$, the inseparability locus of the pentahedral quintic, and the singular-output locus are proper closed conditions on the source, provided one is not in a degenerate component of the construction. Therefore rational points avoiding them are Zariski dense. Exhaustive primitive enumeration must eventually meet this open set. The argument gives no useful universal height bound. □

13 Certification and debugging checks

A robust implementation should perform the following checks.

13.1 Universal checks

1. Verify that the stored matrices for Pinkham's six-dimensional representation satisfy the Coxeter relations for $W(E_6)$.
2. Verify that the 27-line permutation action agrees with the action on the Picard lattice and that q_{27} is $W(E_6)$-equivariant:
$$M_w q_{27} = q_{27} P_w.$$
3. Verify the normalized gamma equivariance
$$\widehat{\Gamma}(M_w t) = \widehat{P}_w \widehat{\Gamma}(t)$$
for the chosen generators and several random rational regular vectors t.
4. Verify that the power-sum formulas (8) are invariant under the stored $W(E_6)$ action.
5. Verify that the implementation sums over forty gamma labels, not over eighty signed labels.

13.2 Instance checks

1. Verify that the labelled Galois group of f has the declared order and acts on the roots in the declared way.
2. If the root algebra is used, verify that the chosen intertwiner
$$\mathbf{Q}^{\Omega_f} \to V$$
has rank 6 and satisfies (28) for the generators of G.
3. If the 27-line algebra is used, verify that the orbit resolvents have the expected degrees and are separable.
4. Verify that the descended idempotent satisfies
$$e_\rho^2 = e_\rho$$
and that
$$\dim_{\mathbf{Q}} \operatorname{im}(e_\rho) = 6.$$
5. For candidates, verify that
$$\Delta_{E_6,\rho}(a) \neq 0$$
before evaluating the higher-degree Clebsch functions.
6. Verify that the computed Clebsch vector is not the zero weighted point and that $E \neq 0$ before constructing the pentahedral quintic.
7. Verify separability of the pentahedral quintic and smoothness of the final cubic surface.
8. As an independent final certification, compute the 27 lines of the output surface and compare the resulting Galois action with the prescribed representation ρ.

14 A staged implementation plan

The full resolvent implementation is the clean mathematical endpoint, but it need not be implemented all at once.

14.1 Stage 1: remove B_ρ

Keep the existing full-splitting-field construction of the six-dimensional matrix D_ρ, but delete the ten-dimensional point-search machinery. For each primitive vector $u \in \mathbf{Z}^6$, form

$$t = D_\rho u \in L^6,$$

check $\delta_{E_6}(t) \neq 0$, evaluate the normalized gamma formulas, compute A, B, C, D, E, coerce these invariant values to $\mathbf{Q}$, and run the pentahedral trace construction. This tests the idea that the ten-dimensional descent is unnecessary.

14.2 Stage 2: try the root algebra

Before building any 27-line resolvents, test whether the root permutation module of f contains V as a quotient. If so, the algebra

$$\mathbf{Q}[x]/(f)$$

may already support the source descent. This is the cheapest possible no-full-L model.

14.3 Stage 3: use the 27-line resolvent

If the root algebra does not contain V, construct the 27-line resolvent algebra and the descended idempotent e_ρ. This gives a uniform degree-27 model of the twisted source. The main technical work is implementing the descended linear maps and invariant evaluators without losing the relative labelling of orbit factors.

Practical expectation. Stage 1 should already be faster than the original ten-dimensional search. Stage 2 can be much faster when the given polynomial's root action already contains the six-dimensional representation. Stage 3 is the robust general method for avoiding a full explicit $\mathbf{Q}$-basis of the splitting field.

15 What remains expensive

The resolvent approach removes the need to do linear algebra in a vector space of dimension $[L : \mathbf{Q}] = |G|$, but it does not make the arithmetic trivial. The following steps may still be nontrivial.

- Constructing resolvent polynomials can be expensive if the chosen resolvent has large degree, large coefficients, or difficult stabilizer.
- If Ω has several orbits, the descended idempotent and invariant functions require correspondences between orbit factors, not merely independent polynomial fields.

- The Clebsch functions have high degree in the source coordinates. Their values may have large numerators and denominators. Modular evaluation and rational reconstruction may be preferable to direct symbolic expansion.
- The algorithm gives a dense supply of rational points but no small height bound. LLL reduction and good source lattices remain important.

These are implementation costs inside degree d or degree 27 étale algebras, not costs inside the full splitting field of degree $|G|$.

16 Minimal final form of the algorithm

The mathematically compressed version is the following.

A1. From f and the labelled group $G \le S_{\deg f}$, regard $\operatorname{Spec} L \to \operatorname{Spec} \mathbf{Q}$ as a G-torsor, without constructing L as a $\mathbf{Q}$-vector space.

A2. Restrict Pinkham's six-dimensional $W(E_6)$ representation along $\rho : G \to W(E_6)$.

A3. Choose a finite G-set Ω and a G-equivariant surjection $\mathbf{Q}^{\Omega} \twoheadrightarrow V$. Use the root set of f if it passes the rank six intertwiner test; otherwise use the 27-line set.

A4. Construct the associated resolvent algebra

$$\mathcal{A}_{\Omega} = (L \otimes \mathbf{Q}^{\Omega})^{G}$$

and the descended idempotent e_ρ whose image is the twisted source V_ρ.

A5. Enumerate rational points of

$$V_\rho = \operatorname{im}(e_\rho).$$

A6. For each candidate, evaluate the descended invariant functions

$$\Delta_{E_6,\rho}, \quad A_\rho, B_\rho, C_\rho, D_\rho, E_\rho.$$

Reject the candidate if it lies on the boundary or outside the valid pentahedral locus.

A7. Convert the Clebsch vector to a pentahedral quintic and then to a cubic form by the trace construction.

A8. Check smoothness and, for certification, verify the induced Galois action on the 27 lines.

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